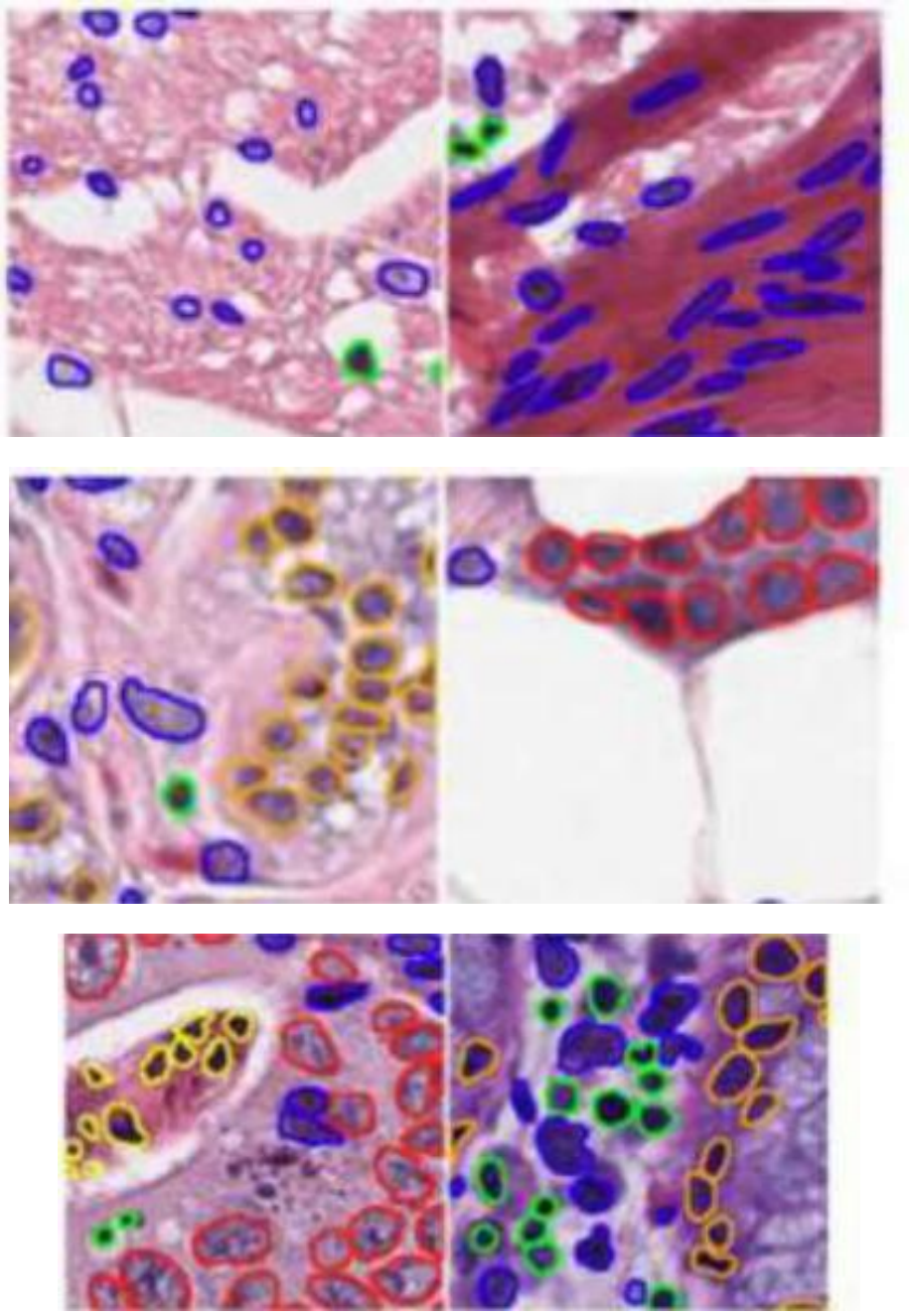
BIOMEDICAL DATA DESIGN I

Nuclei instance segmentation across diverse tissues

HoVer-Net, CellViT-UNI, and PM-Net

Weekly report • October 15, 2026

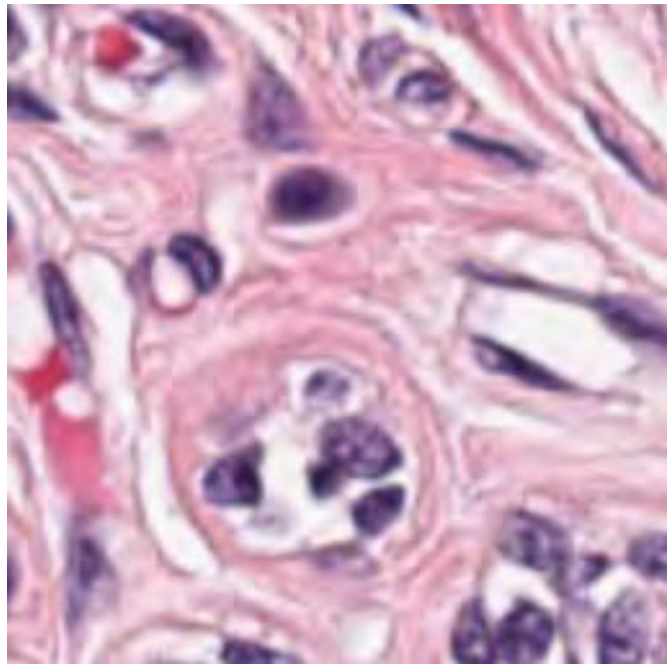
Xinhao Jin



THE TASK

A nucleus needs its own mask and class label

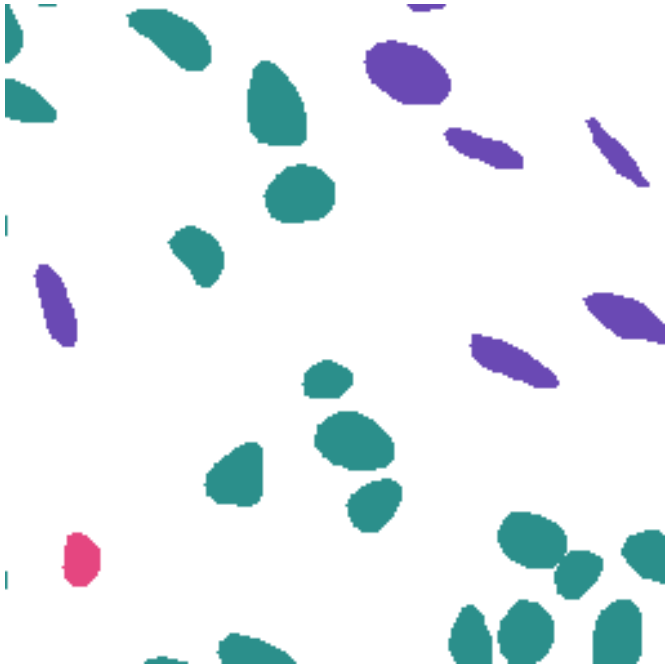
H&E image



Individual instances



Five nucleus classes

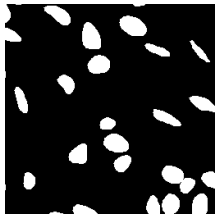


Core question: does one model work consistently across tissues and classes?

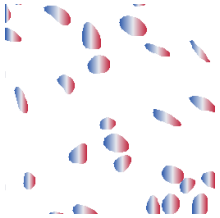
One shared encoder feeds three decoder branches

CNN encoder ResNet-50

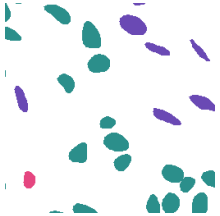
NP: nucleus pixels



HV: relative position



TP: nucleus type



Separate jobs,
shared image features

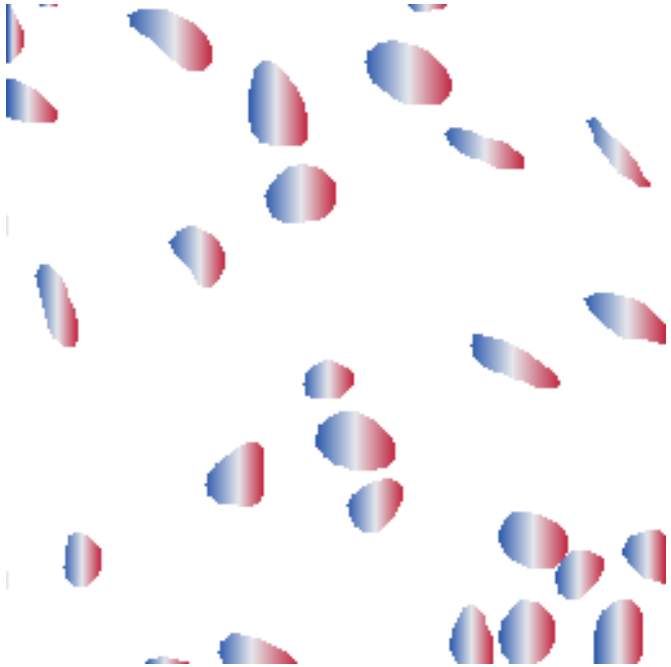
NP and TP are pixel classification. HV predicts two continuous values.

Each nucleus has its own horizontal and vertical coordinates

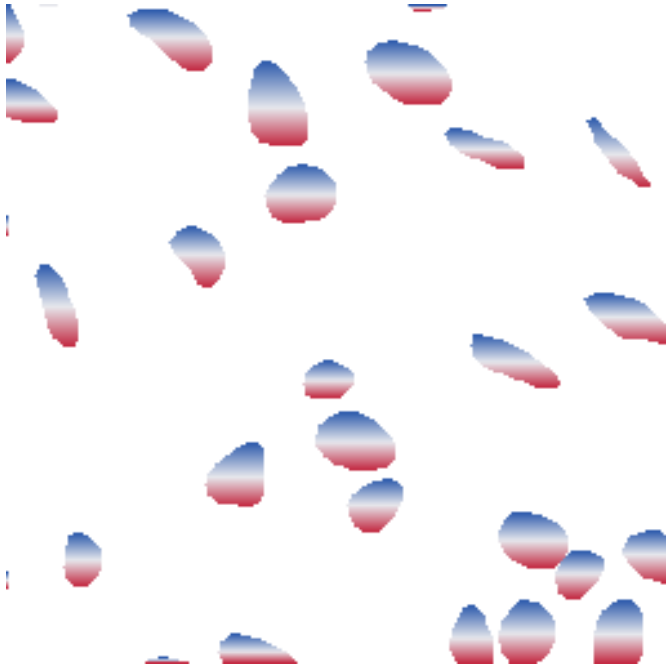
Instance IDs



Horizontal offset

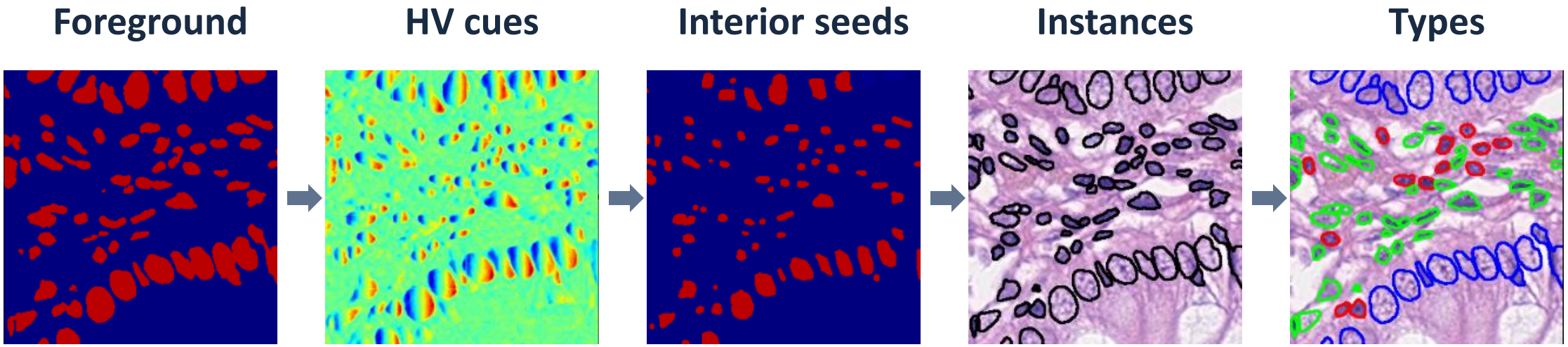


Vertical offset



Blue = negative offset. Red = positive offset. White background is outside nuclei.

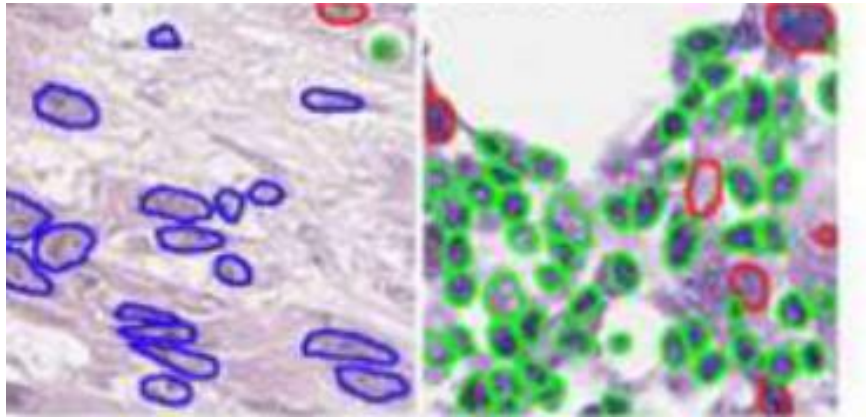
Watershed turns pixel maps into separate objects

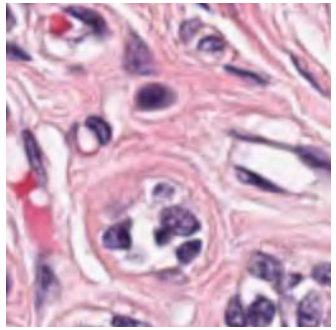


Original prediction and decoding panels from Graham et al. (2019), Fig. 1.

Missing seeds can merge nuclei. Extra seeds can split one nucleus.

UNI learns pathology features before our task training





More than 100 million image patches

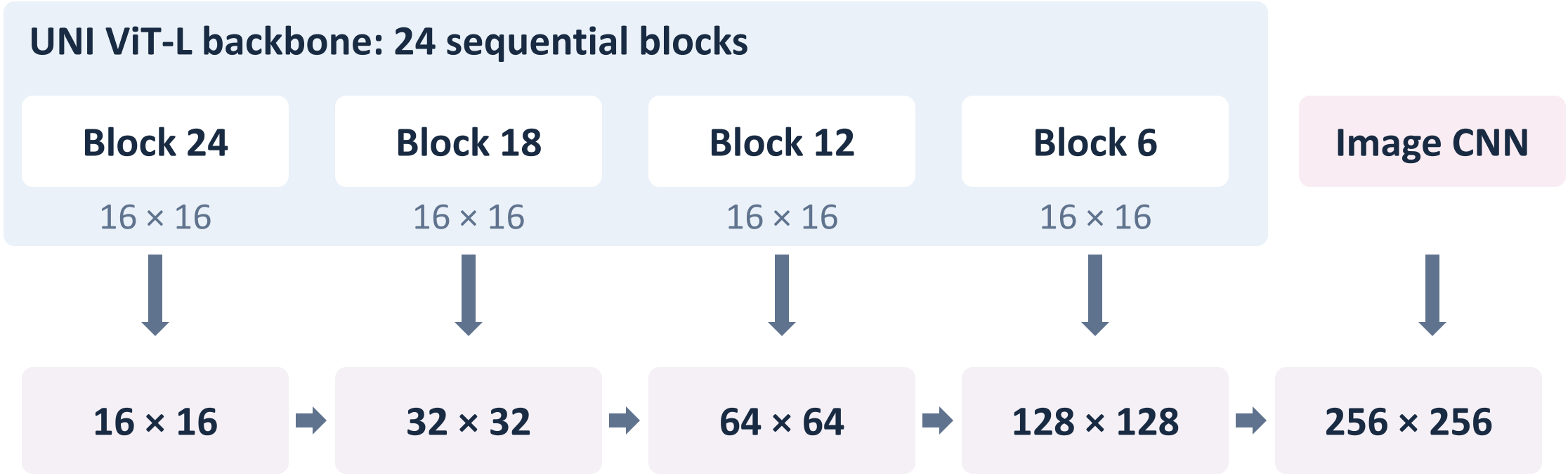
Over 100,000 H&E whole-slide images

Self-supervised pretraining
across 20 major tissue types

UNI supplies image features. The nucleus decoder still has to learn our task.

CELLVIT DECODER

Features from four depths help recover small structures



Independent depth taps above. Decoder resolution increases from left to right.

CellViT retains NP / HV / TP heads and watershed instance decoding.

BASELINE EVIDENCE

mPQ improves substantially while binary PQ barely moves

Own training, no TTA	mPQ	bPQ
HoVer-Net	0.4564	0.6613
CellViT-UNI	0.4995	0.6653

ΔmPQ +0.0431 **ΔbPQ +0.0040**

38.5% **of Dead nuclei had almost no overlap with any prediction**

Three-split means above. The miss-rate diagnosis is a separate historical analysis.

What the completed studies tell us

Route	What we tested	Result and interpretation
Typing and data	New typing heads, copied nuclei, generated images	No stable gain. Changing a class cannot recover a missed nucleus.
Models and scale	Alternative models and larger input images	No overall win. Larger images recover nuclei but also add errors.
Mechanism checks	HV targets, class conflict, label agreement	No single cause established. Annotation uncertainty remains.
Cost and supervision	Filtered extra masks and crop supervision	Small scale gain is costly. The supervision fix fails its criteria.

No completed route has shown a clear overall improvement.

Filtered scale additions keep a small Dead gain at high cost

Nine matched split / seed pairs	mPQ	bPQ	Dead PQ
Matched baseline	0.49760	0.66530	0.17255
P2 additions	0.49769	0.66443	0.17997
EF-P2 filtered additions	0.49828	0.66529	0.17954

+0.0070 Dead PQ

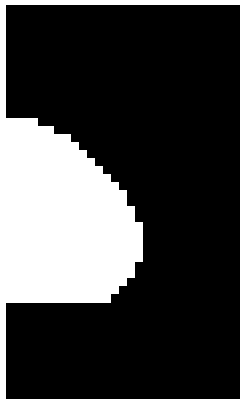
≈6.1× inference cost

Image-bootstrap intervals span zero. Historical test folds were reused across rounds.

Does changing HV targets repair crop-related misses?



Visible nucleus



Cropped mask

E4: cropped Dead nuclei were missed more often.

Arm	Crop prob.	HV minimum area
A0	0.1	30 pixels
A1	0.5	30 pixels
A2	0.5	1 for new small crop fragments
A3	0.5	1 for all objects

A2 – A1 isolates selective HV supervision from stronger cropping.

The supervision fix fails the development criteria

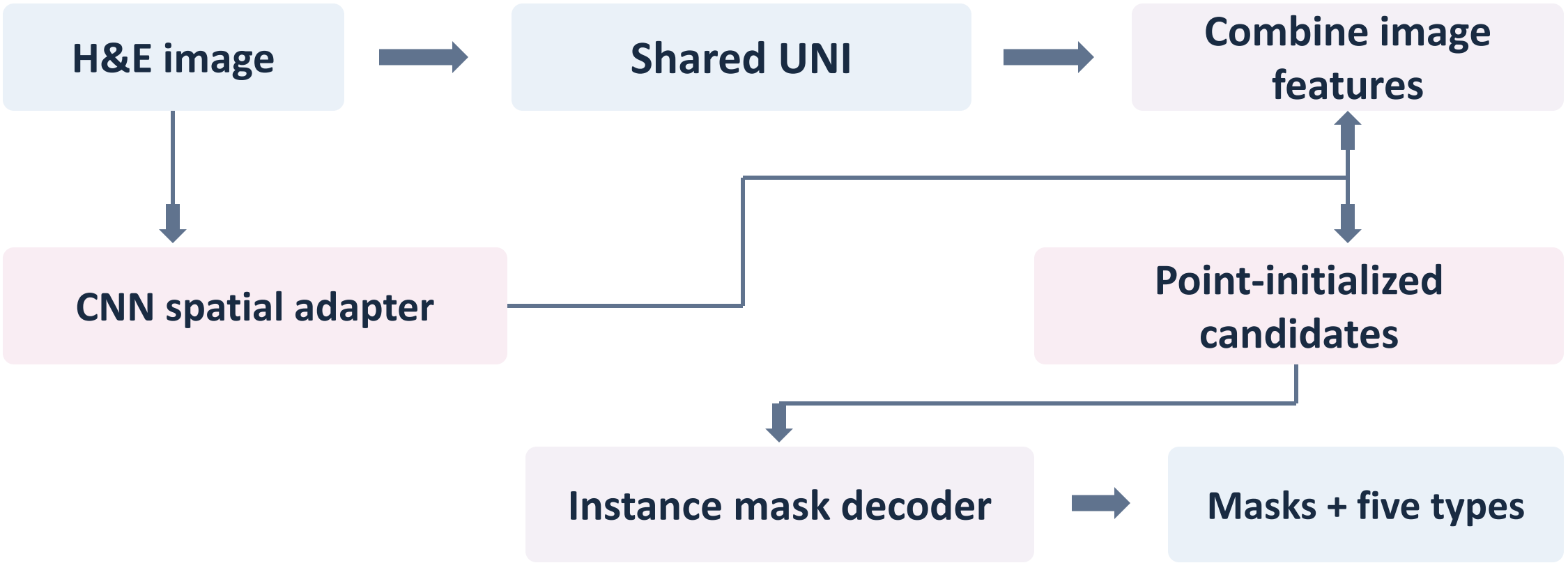
Mean change, two seeds	mPQ	bPQ	Dead PQ	Strict Dead
A1 − A0	−0.0053	−0.0061	+0.0376	+0.0088
A2 − A0	−0.0063	−0.0054	+0.0246	−0.0009
A2 − A1	−0.0010	+0.0007	−0.0131	−0.0098

Selective HV adds no Dead benefit over crop-only training.

A1 strict Dead rises +0.0088. A2 falls versus A1 on both Dead endpoints.

PM-NET ARCHITECTURE

Each candidate nucleus predicts its own mask

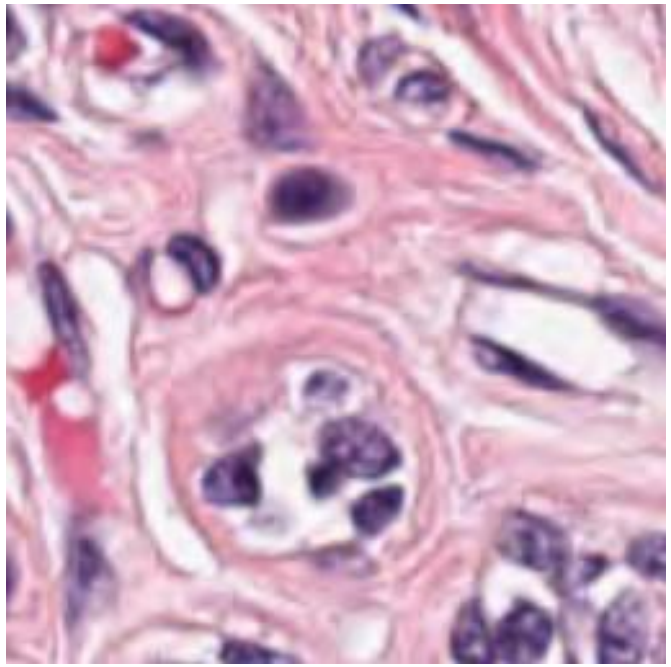


Proposed: find interior points, predict masks, resolve overlaps.

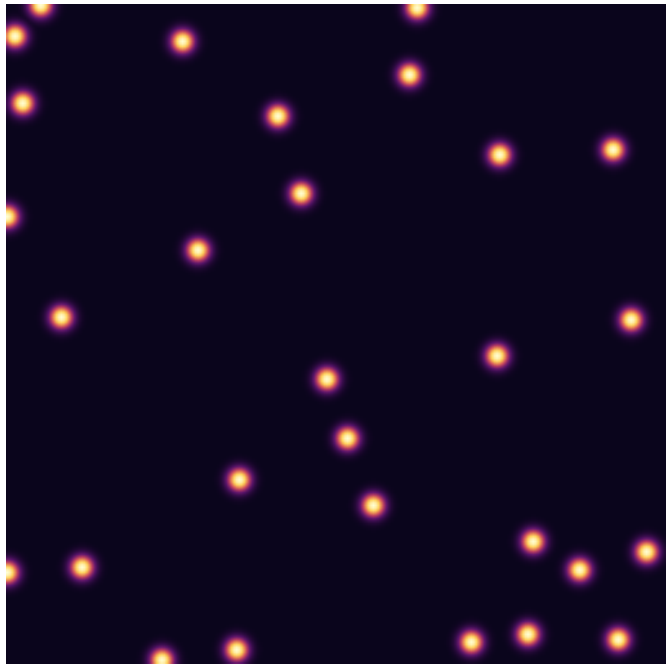
POINT SUPERVISION

A point target must stay inside the visible nucleus

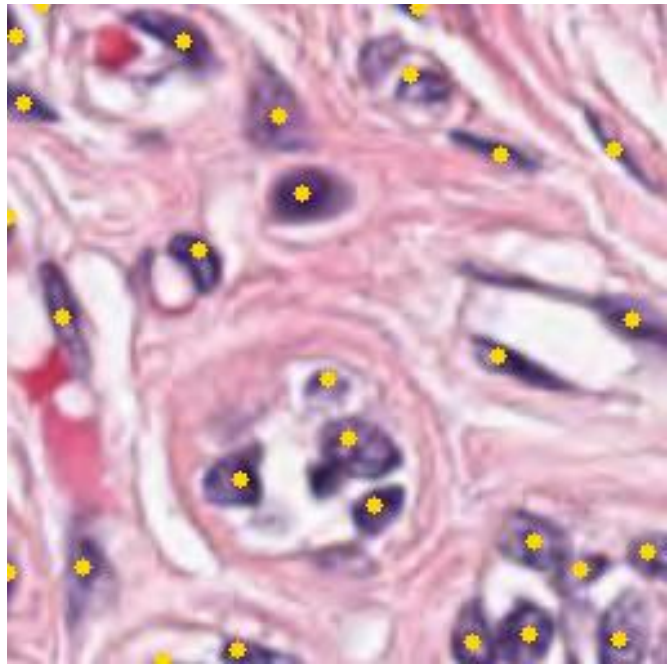
Image



Deepest interior points



Inside-mask points



Proposed target: the visible pixel farthest from the boundary.

Controlled tests separate useful design from extra capacity

Comparison	Question
Candidates with vs without interior-point hints	Do the point hints help?
Simple vs gated mixing of image features	Does the learned gate help?
A simpler adapter with the same parameter count	Is the gain only extra capacity?

276 nuclei in one patch: test 512 candidates

Each candidate reads a few image locations to limit cost

512 candidates still need a complete memory and speed check.

EVALUATION PLAN

Detection gains must preserve quality and inference cost

Endpoint	Proposed development requirement
Interior Dead recall	Mean gain ≥ 0.03; paired lower 95% bound > 0
Dead PQ	Mean gain ≥ 0.005; both seeds improve
mPQ / bPQ	Per-seed loss no more than 0.002
Strict scores and instance errors	No strict decline; mean merge/split rates do not rise
End-to-end latency	At most 2× the single-scale baseline

Matched UNI initialization, schedule, seeds, final checkpoints, and evaluator.

Tissue and class performance remains uneven

Tissue	mPQ	Common-class PQ	Unique test images
Lung	0.4064	0.3780	131
Skin	0.4264	0.4140	100
Colon	0.4370	0.4217	972
Breast	0.5155	0.4891	1,602

Lung, Skin, and Colon rank in the bottom five in all three splits.

Nine frozen split/seed pairs. Common classes: neoplastic, inflammatory, connective.

Focus: diagnosed failures across tissues. Unseen-tissue transfer is untested.

Detection, mask quality, and type all affect the score

mPQ	**Type-aware quality**	PQ averaged across classes and tissue types
bPQ	**Instance quality**	All nuclei treated as one foreground class
Dead PQ	**Rare-class quality**	Focus on dead nuclei
Strict scores	**Absent-class penalties**	Retain false positives when a class is absent

Transfer to an unseen tissue requires excluding it from training and tuning.

HV area thresholds and class conflict are incomplete explanations

E2a: scale × HV target support, four arms and two seeds

Lowering the x1 cutoff reproduces only 9.6% of the x2 Dead gain.

DSB: test the need for a separate Dead expert

Mean gradient cosine +0.749; none of 64 valid batches is negative.

Decision

Keep HV cutoff 30. Stop the specialist route before expert training.

The measured class-conflict premise failed under the tested conditions.

Detector agreement is useful evidence, but not ground truth

E1-C / E1-T: agreement across detectors

66–73% candidate support; 72.5% vote for the base model's disputed type.

M: merge inflammatory and connective labels

Official mPQ changes by +0.0519 in a different label space.

E3: learn candidate utility

AUC 0.721 / 0.737, but strict Dead criteria fail. Identity fallback.

Expert review remains unavailable. Five-class method scores are unchanged by M.

Selective HV supervision does not reliably reduce misses

−0.0086

Mean reduction in miss rate, A1 − A2

95% interval: [−0.0409, +0.0177]

116 eligible Dead objects in 38 images; inference covered 2,523 fixed windows.

This recipe fails the frozen development criteria.

APPENDIX A5: RARE-CLASS SCOPE

Dead gains have limited tissue coverage

2,908 Dead nuclei in 232 positive images

59.5% of Dead nuclei are in Lung. Seven tissues have no Dead labels.

EF-P2 change in Dead PQ	Mean change	Image-bootstrap 95% interval
Lung	+0.0208	[+0.0115, +0.0309]
Colon	+0.0093	[−0.0005, +0.0191]
Stomach	−0.0014	[−0.0031, −0.0001]

Pointwise exploratory intervals, conditional on trained seeds. Historical test reuse.

Dead detection is a valid target. Broad tissue gains remain unproven.